

Numerical of Motion in straight line: The 'UTSAV' Concept
3 equations of motions

- 1 $v = u + at$
- 2 $s = ut + \frac{1}{2}at^2$
- 3 $v^2 - u^2 = 2as$

Horizontal motion

- u = initial velocity
- t = time
- s = distance
- a = acceleration
- v = final velocity

Vertical motion

$a = g$ → Acceleration due to gravity

Free fall → $u = 0$

Initial speed

- $v = u + gt$
- $h = ut + \frac{1}{2}gt^2$
- $v^2 - u^2 = 2gh$

Against the gravity ($a = -g$)

- $v = u - gt$
- $h = ut - \frac{1}{2}gt^2$
- $v^2 - u^2 = -2gh$

Final point = 0

Final velocity = 0

motion of a body moving with speed along the circular path

Centripetal acceleration

Acceleration towards the centre in circular path

$a_c = \frac{v^2}{r}$

Change in velocity at every point

Relative change in position with respect to reference point

Straight line motion / Rectilinear motion

Uniform Circular Motion



physical quantities

- Distance → scalar quantity (only Value)
- Displacement → vector quantity (Value + Direction)

Types

- Uniform Motion → Equal distance covered in equal intervals
- Non-Uniform Motion → Equal distance in unequal intervals

Speed

Rate of change of motion

$Speed = \frac{Distance}{Time}$

$\xrightarrow{m/km}$
 $\xrightarrow{s/hr}$

S.I Unit - m/s

scalar quantity

Velocity

$Velocity = \frac{Displacement}{Time}$

Unit: m/s

vector quantity

Change in velocity = Final speed - Initial speed

Acceleration

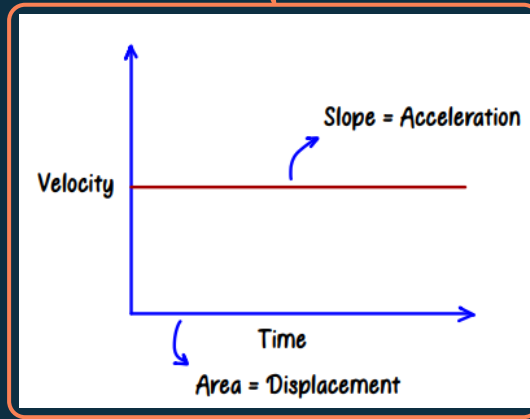
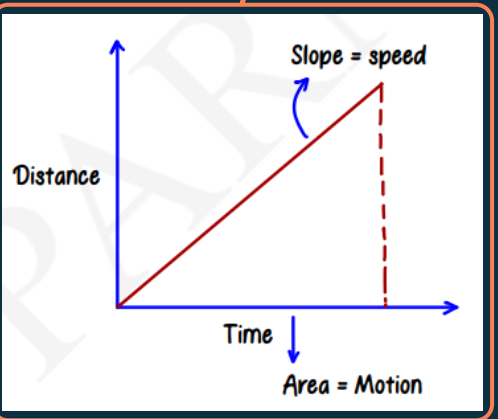
Rate of change in velocity

vector quantity

$Acceleration = \frac{m/s}{s} = m/s^2$

$\xrightarrow{a_c}$

Graph

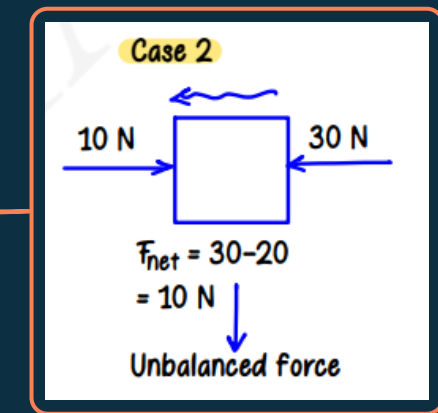
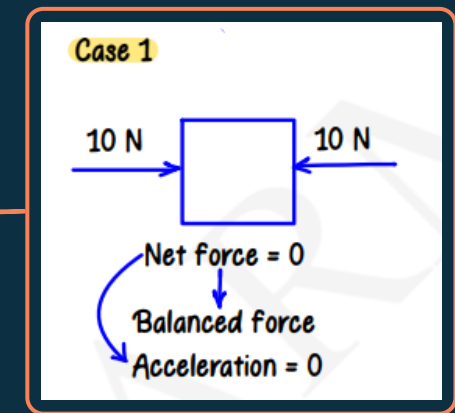


PARMAR SSC Laws of Motion

Force

- produces change in velocity
- S.I unit - Newton (N)
- Types
 - Contact
 - eg: Frictional, Spring, Muscular etc
 - Non-contact
 - eg: Magnetic, Gravitational, Electrostatic etc
 - Conservative
 - Work done is independent of the path
 - Work done is 0 in case of closed path
 - Eg: non-contact force
 - Exception - spring force
 - Non-conservative
 - Work done is dependent on path taken
 - Eg: frictional force

concept

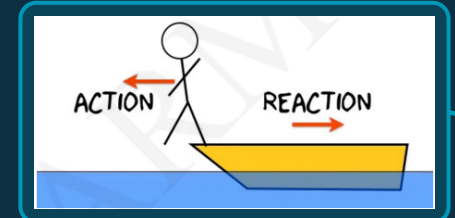


1st Law

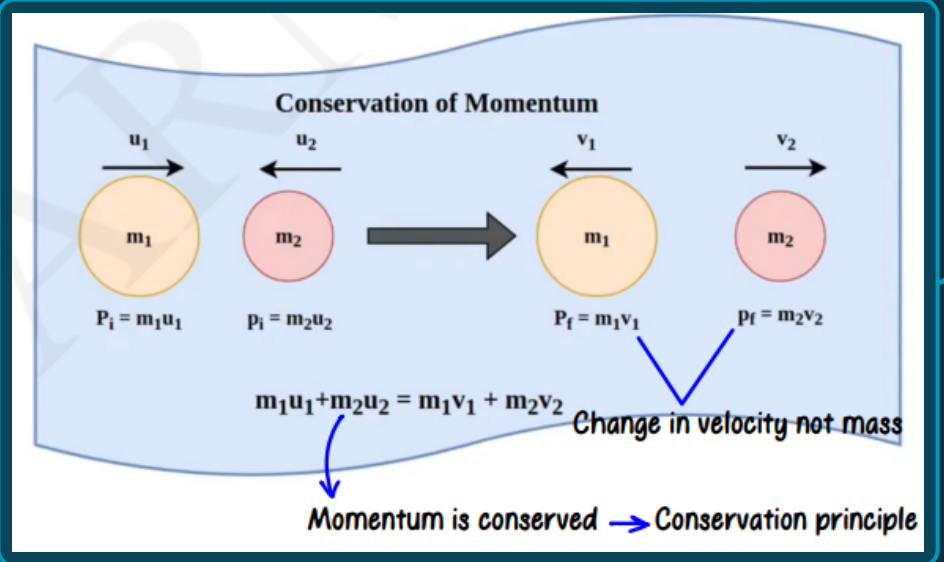
- An object at rest remains at rest or in motion remains in motion at a constant velocity unless an external force is applied on it.
- Inertia
 - Change state of object
 - Inertia $\propto$ Mass
- Examples
 - Dry leaves and fruits falls when we shake a tree
 - A person sitting in a moving car bends forward when the car stops

3st Law

For every action, there is an equal and opposite reaction.

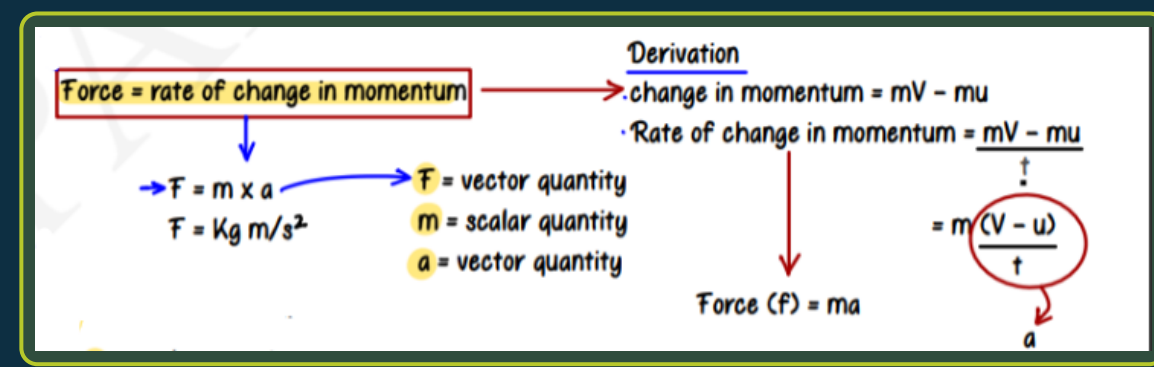
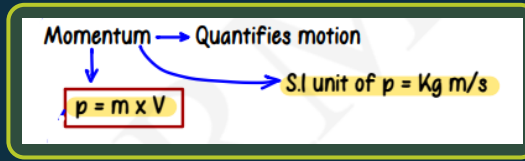


Rocket works on conservation of momentum & Newton's 3rd law



2st Law

Acceleration of an object is directly proportional to the Net Force acting on it and inversely proportional to its Mass.





Universal Law of Gravitation

Every object attracts another object in the universe with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centers.

Formula

$$F \propto \frac{q_1 q_2}{r^2}$$
$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

• q = charge
• S.I unit of charge: Coulomb

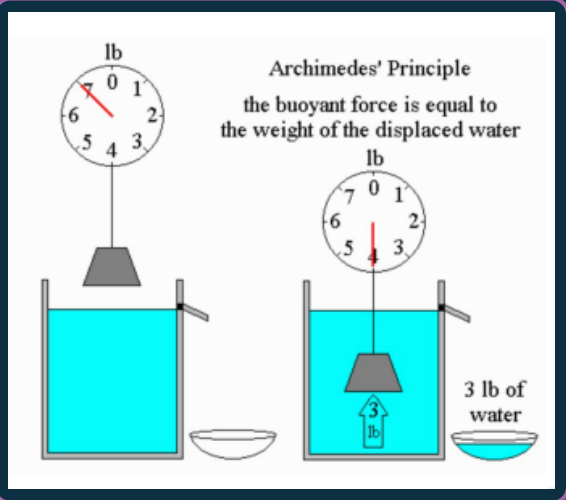
$$F \propto \frac{M_1 M_2}{r^2}$$
$$F = \frac{G M_1 M_2}{r^2}$$

Gravitational Constant

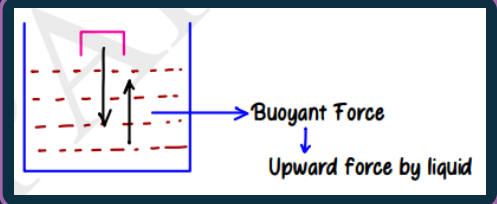
• G = Gravitational constant
• Value: $6.67 \times 10^{-11} \text{ Nm}^2/\text{Kg}^2$ S.I unit
• Discovered by: Henry Cavendish (1798)

Variation in gravity

- Gravity is more in poles than the equator
- Gravity decreases with altitude
- Gravity in moon is 1/6th of Earth's gravity



Archimedes Principle



Less buoyant force
River water has less density → No salt content

Sea water has high density → High salt content
High buoyant force

Weight

$$\text{Weight} = m \times g$$

Constant Gravity differs

force by which Earth attracts a mass

Relative Density

$$R.D = \frac{\text{Density of a substance}}{\text{Density of water}}$$

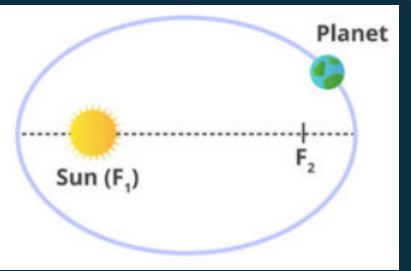
No unit

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

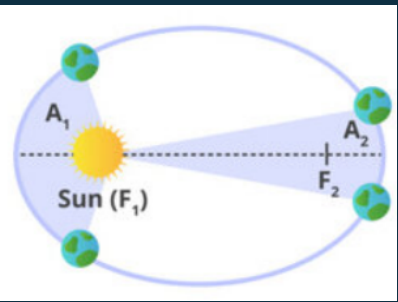
S.I unit: $\frac{\text{Kg}}{\text{m}^3}$

Kepler's Laws

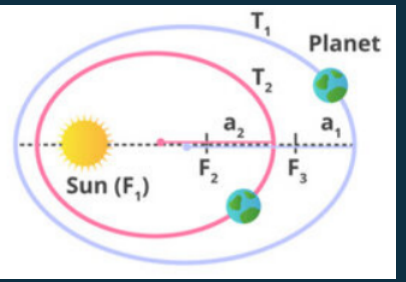
Law of Orbit - Planets move in elliptical orbits with the Sun as a focus

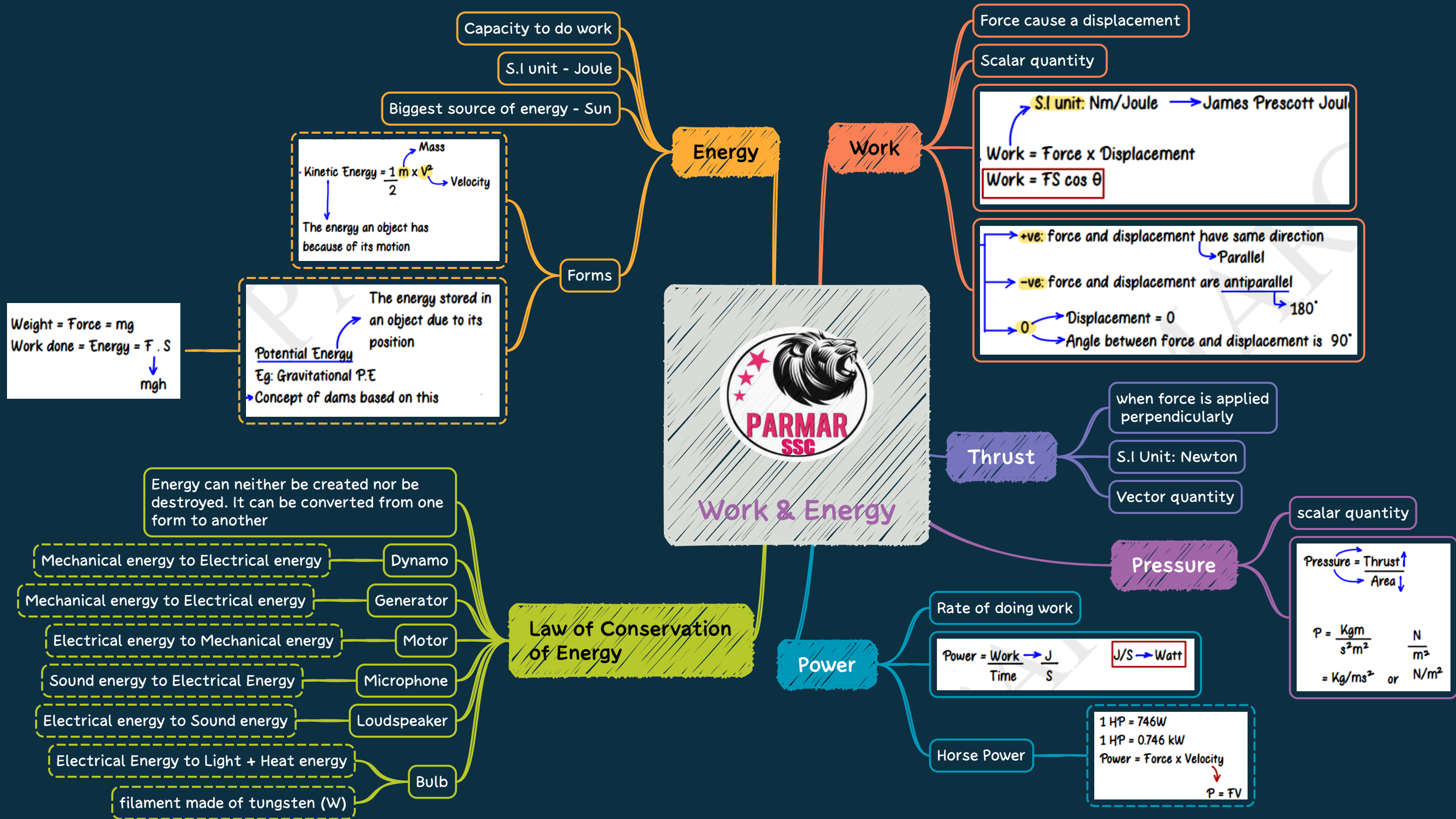


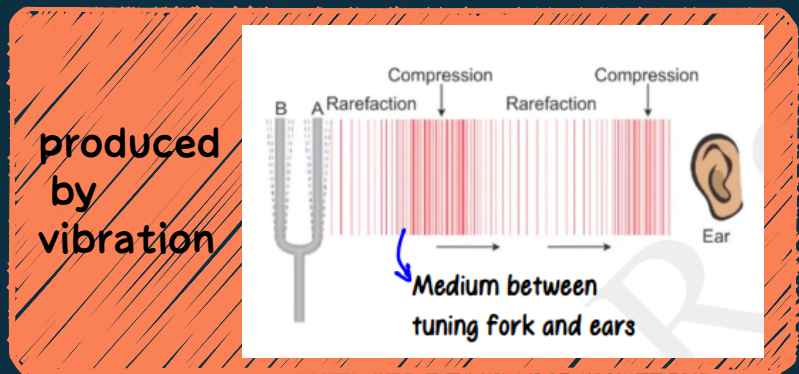
Law of Orbit - The line joining the planet and the Sun sweep out equal areas in equal interval of time



Law of Time Period - Cube of mean distance of a planet from the Sun is proportional to the square of the time period/orbital speed







Sound

Waves

Mechanical wave

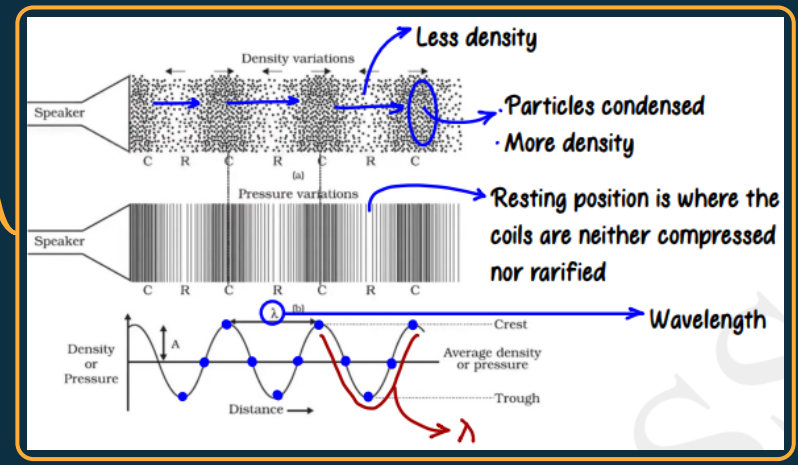
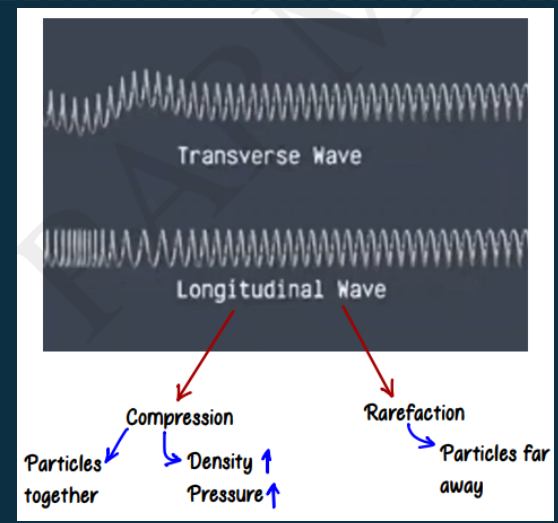
Medium is required to propagate

Ex: Sound waves

Non-mechanical wave

No medium is required to propagate

Ex: Light waves



• Solid > Liquid > Gas
• Light = 3×10^8 m/s
• Speed of light is more than the speed of the sound

SONAR

Device that is used for detecting & locating objects specially underwater by the means of sound waves sent out to be reflected by the objects

Sound Navigation and Ranging

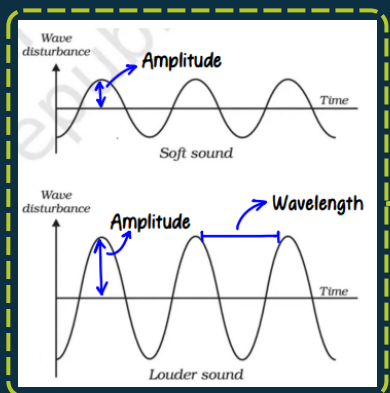
Ultrasonic sounds

- Produced by bats, dolphins
- Penetrating power is very high
- banned in gender determination
- to monitor growth & development of fetus
- to identify kidney stones & to break it
- to identify the conditions of our organs

Frequency (γ) = $\frac{1}{\text{Time}}$ → Unit: s^{-1} ; Hertz
• Determines pitch of a sound
• Girls have high pitch, high shrillness → Vocal cords vibrates quickly

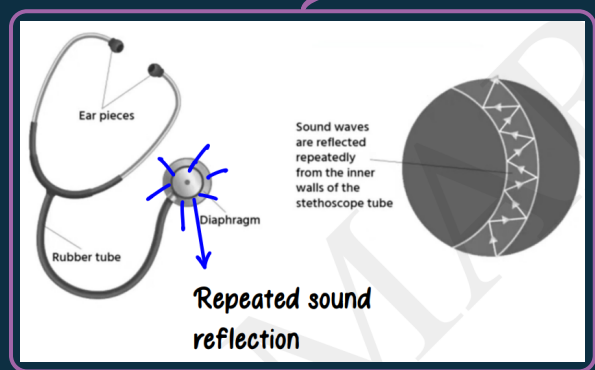
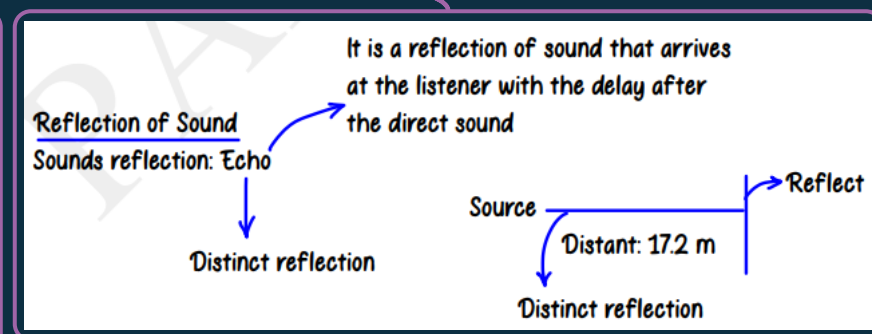
Characteristics

Amplitude: determines Loudness → Unit: dB (decibel)
• Can be defined as the loudness of the amount of maximum displacement of vibrating particles of the medium from their mean position when the sound is produced
Limit: 0-130 dB (sound above this is considered as noise)



Echo

Speed of sound



Speed of Sound in Different Media at 25 °C		
State	Substance	Speed in m/s
Solids	Aluminium →	6420
	Nickel	6040
	Steel	5960
	Iron	5950
	Brass	4700
	Glass (Flint)	3980
Liquids	Water (Sea)	1531
	Water (Distilled)	1498
	Ethanol	1207
	Methanol	1103
Gases	Hydrogen	1284
	Helium	965
	* Air	346
	Oxygen	316
	Sulphur dioxide	213

Audible range

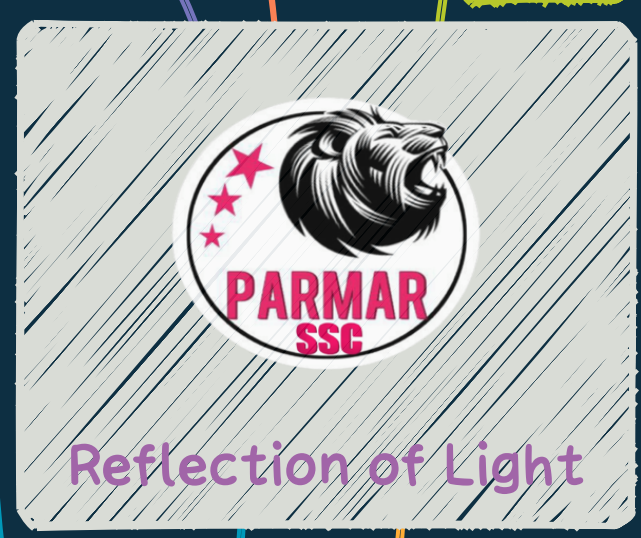
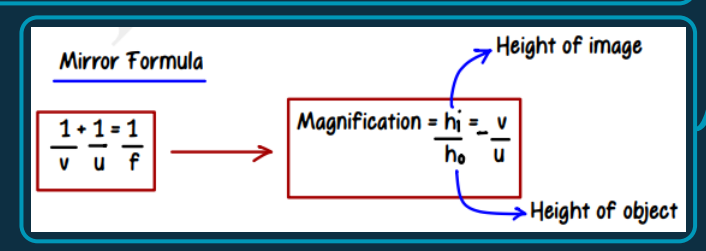
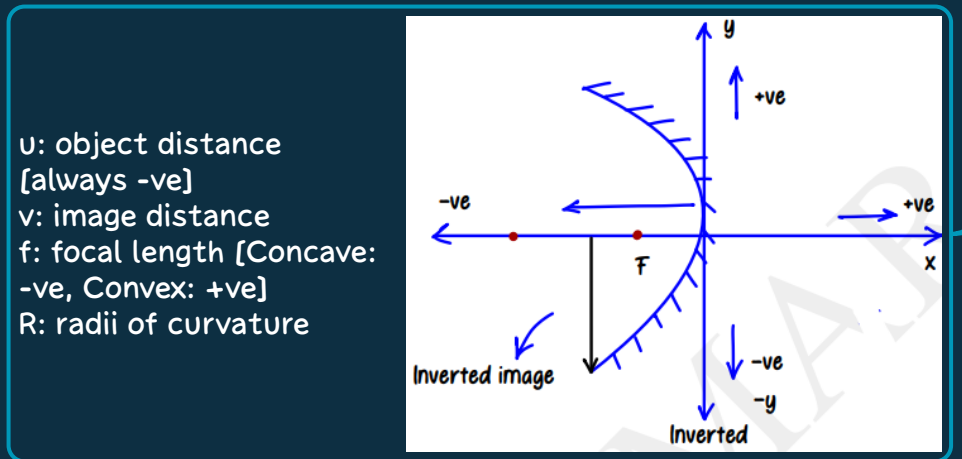
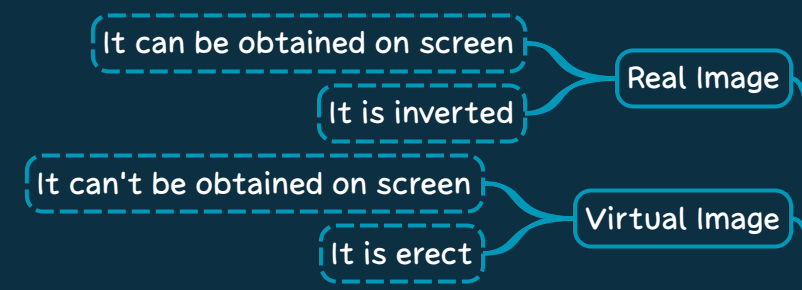
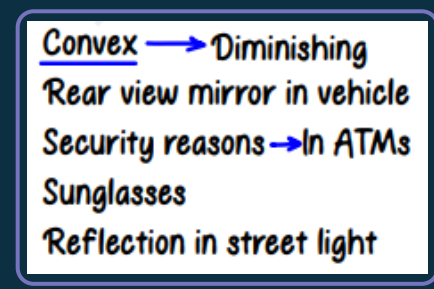
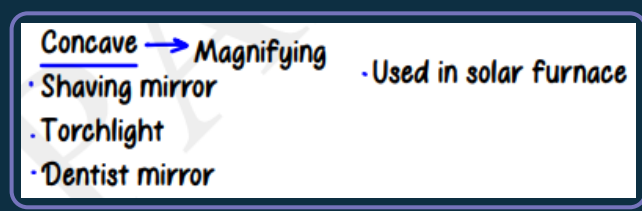
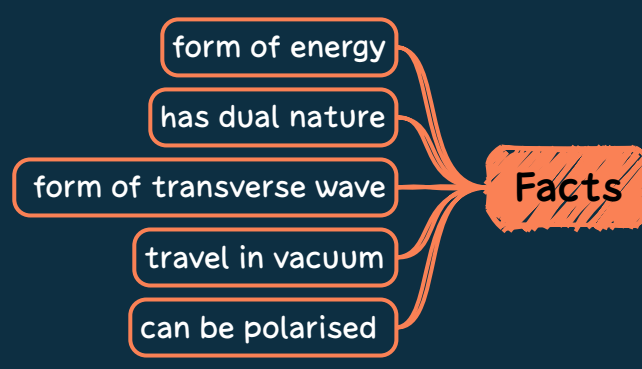
- 20Hz - 20,000Hz
- Infrasonic sound - Less than 20Hz
- Ultrasonic sound - More than 20,000Hz

- Subsonic: Mach < 1
- Supersonic: >1; < 5
- Hypersonic: Mach > 5
- Transonic: Mach = 1

Mach no = $\frac{\text{Speed of any body}}{\text{Speed of sound in that medium}}$

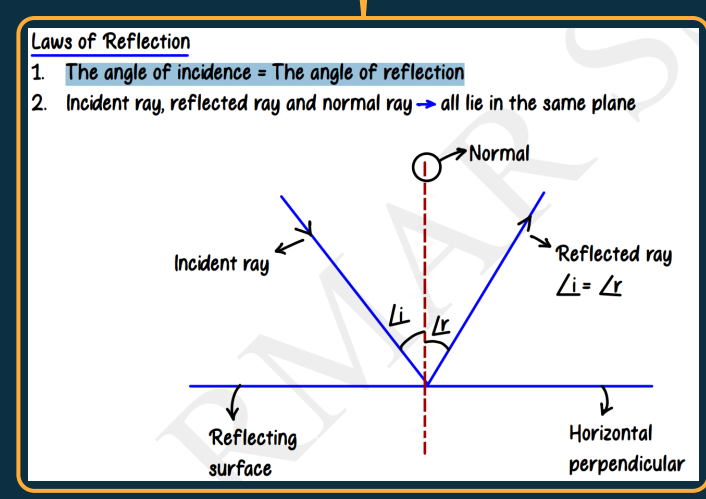
repeated reflection of sound





Image

Reflection



Mirror

Plane Mirror

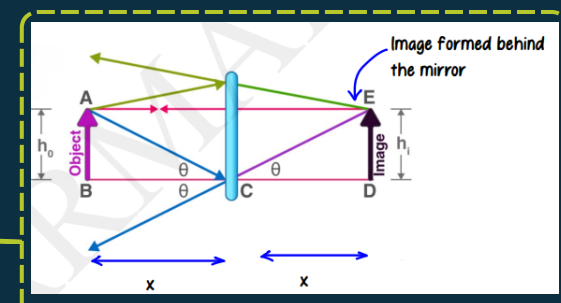
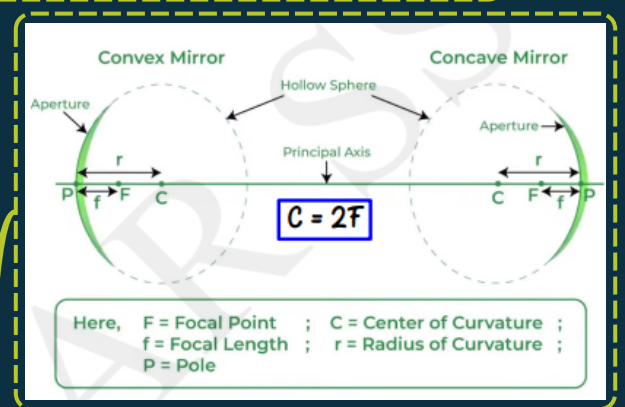


Image distance = Object distance
 Object size = Image size
 Erect Virtual
 Laterally inverted



Spherical Mirror

Concave Mirror

Position of object	Figure	Position of image	Nature of image
1. At infinity		At the principal focus or in the focal plane	Real, inverted, extremely diminished in size
2. Beyond the centre of curvature		Between the principal focus and centre of curvature	Real, inverted and diminished
3. At the centre of curvature		At the centre of curvature	Real, inverted and equal to object
4. Between focus and centre of curvature		Beyond centre of curvature	Real, inverted and bigger than object.
5. At the principal focus		At infinity	Extremely magnified
* 6. Between the pole and principal focus		Behind the mirror	Virtual, erect and magnified

Convex mirror

Position of the object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus F, behind the mirror	Highly diminished, point-sized	Virtual and erect
Between infinity and the pole P of the mirror	Between P and F, behind the mirror	Diminished	Virtual and erect

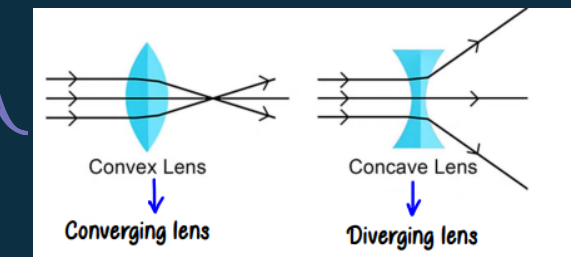
Power = $\frac{1}{\text{Focal Length}}$ $\rightarrow \frac{1}{m} = m^{-1}$ $\rightarrow$ Dioptre (S.I unit of power of lens)

+ve $\rightarrow$ Convex
-ve $\rightarrow$ Concave

Power of Lens

Spherical Lens

- Lens is transparent
- Forms image through refraction



- Incident ray, refracted ray and normal ray $\rightarrow$ all lie in same plane
 - $\frac{\sin i}{\sin r} = \text{constant} \rightarrow$ Given pair of media and light of particular wavelength
- Two mediums

Laws of Refraction

Refractive Index $\rightarrow$ Represented with "n" or " μ "

$n = \frac{\text{Speed of light in air/vacuum}}{\text{Speed of light in given medium}}$

$n_{ap} = \frac{V_p}{V_a}$

Speed of light in air/vacuum: $3 \times 10^8 \text{ m/s}$

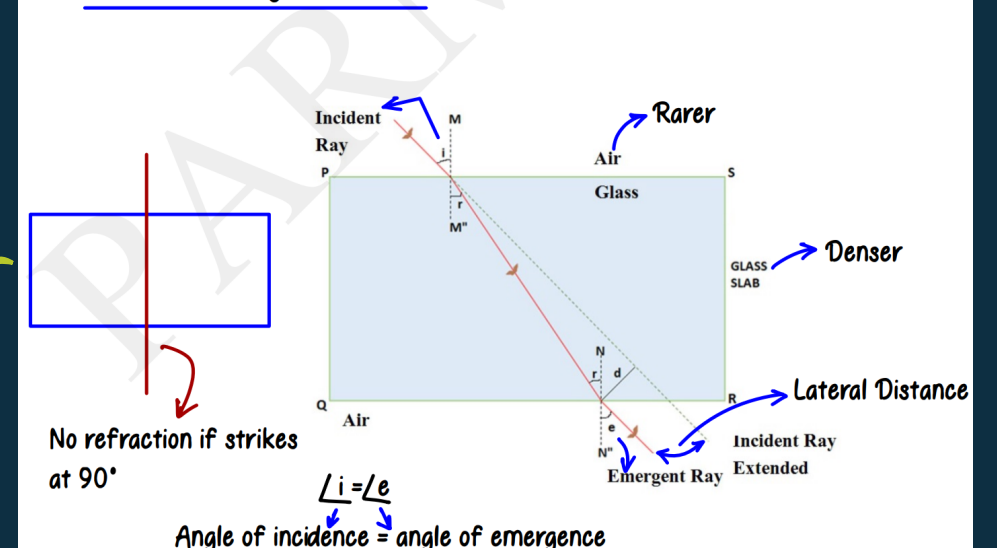
Refractive Index

Material medium	Refractive index	Material medium	Refractive index
Air	1.0003	Canada Balsam	1.53
Ice	1.31	Rock salt	1.54
Water	1.33	Carbon disulphide	1.63
Alcohol	1.36	Dense flint glass	1.65
Kerosene	1.44	Ruby	1.71
Fused quartz	1.46	Sapphire	1.77
Turpentine oil	1.47	Diamond	2.42
Benzene	1.50		
Crown glass	1.52		

Densest medium

Refraction through a Glass Slab

Refraction through a Glass Slab



No refraction if strikes at 90°

$\angle i = \angle e$
Angle of incidence = angle of emergence

Refraction of Light



Image

Medium

Concave Lens
In treatment of Myopia

Convex Lens
In treatment of Hypermetropia
Used as magnifying lens
In camera lens

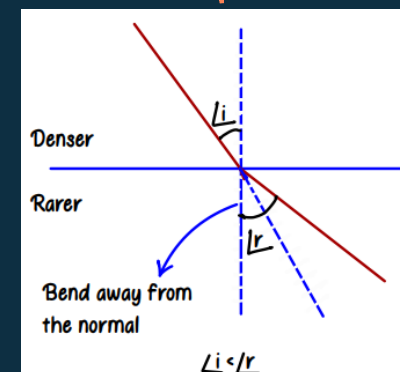
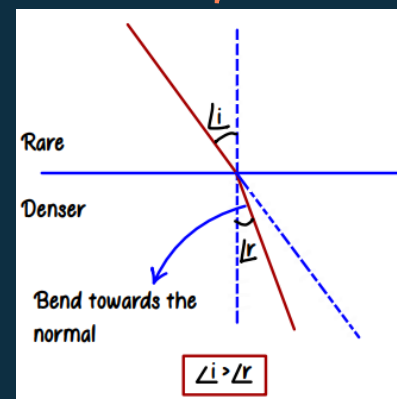
Convex lens

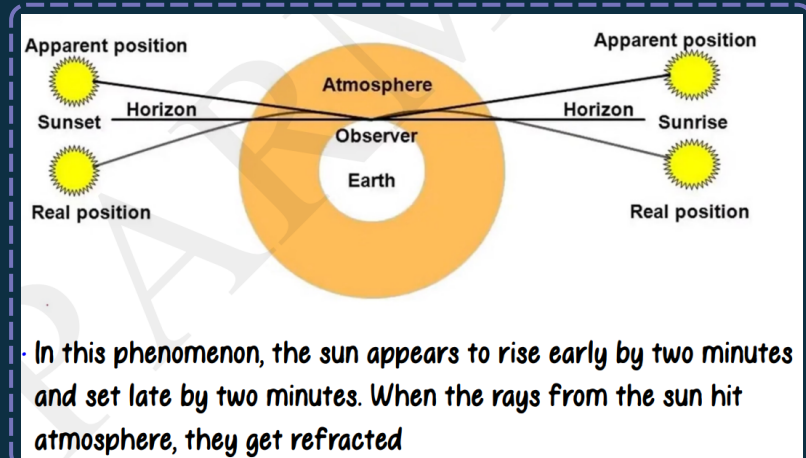
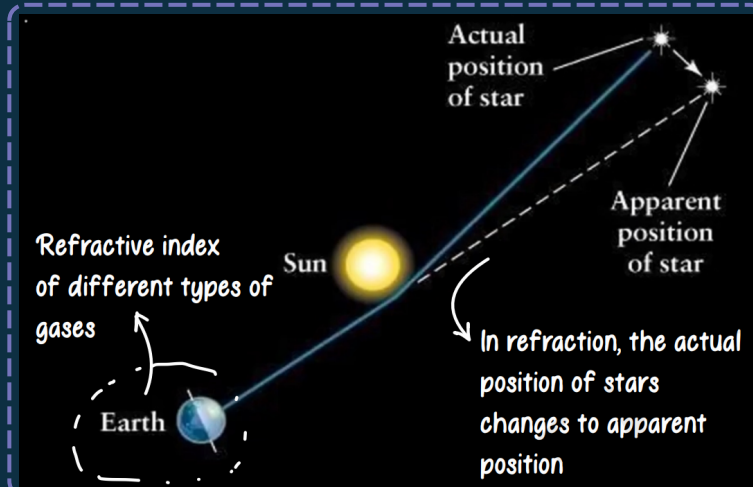
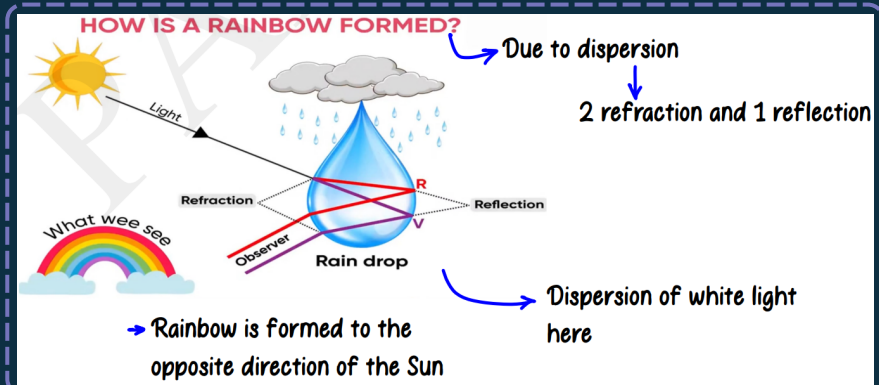
Concave lens

Ray diagram	Position of object	Position of image	Nature of image
(a)	At infinity	At F	Real, inverted and highly diminished
(b)	Between infinity and 2F	Between F and 2F	Real, inverted and diminished

(c)	At 2F	At 2F	Real, inverted and same sized
(d)	Between F and 2F	Beyond 2F	Real, inverted and enlarged
(e)	At F	At infinity	Real, inverted and enlarged
(f)	Between F and O	On the same side of the lens	Virtual, erect and enlarged

Ray diagram	Position of object	Position of image	Nature of image
(a)	At infinity	At F	Virtual, erect and highly diminished
(b)	Between infinity and O	Between F and O	Virtual, erect and diminished





Scattering of Light

blue light is scattered more than the other colours because it travels as shorter, smaller waves

Our sky appears black without atmosphere

red light scatter the least so sunlight travels long path through the atmosphere.

Red colour of Sun during sunrise and sunset

Condition of TIR

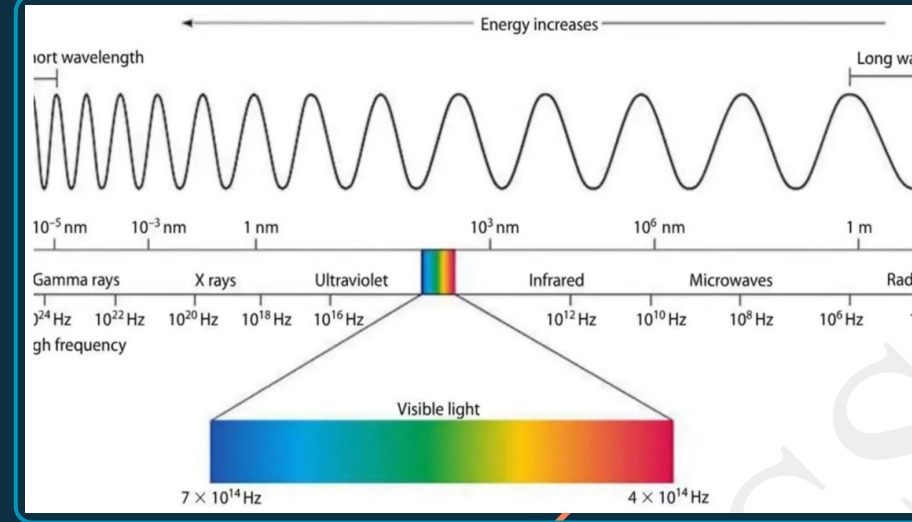
Light should travel from denser to rare medium

light should hit an angle greater than critical angle

This phenomenon is responsible for optical illusions

Ex - Mirage, optical fibers

Total Internal Reflection



Frequency



Facts

Defects of Eye

Myopia / Near Sightedness

- Image formed in front of the retina
- Far object not visible clearly
- Correction: -ve power lens (Concave lens)
- Focal length decreases & Power increases

Hypermetropia / Far Sightedness

- Image formed behind the retina
- Light focuses behind the retina instead of focusing on the retina
- Correction: +ve lens (Convex lens)
- Usually occurs above 40 yrs

Presbyopia

- Lens hardens with age - Loses flexibility
- Age: 55+
- Correction: Concave + Convex lens (Bifocal lens)

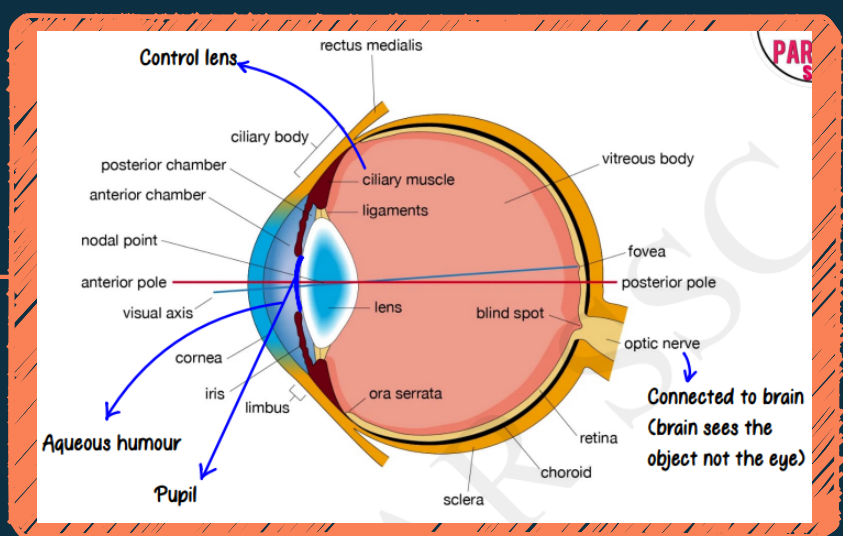
Glaucoma / Trachoma

- Both caused due to increase in eye pressure
- Glaucoma is hereditary
- Trachoma is bacterial infection
- Tonometry to measure your eye pressure

Not curable

Colour Blindness

- It is hereditary
- Retina made of cone cells and rod cells are absent



Cornea

- Outermost part
- Causes refraction of light
- Used in eye donation

Aqueous humour

- Provides nourishment to cornea
- Maintains eye pressure

Iris

- Dark muscular structure
- Controls the size of pupil
- It also determines colour of the eye

Pupil

- To control the amount of light entering the eye

Lens

- focuses the light ray on the retina

Retina

- It is the spot where image is formed
- Here optical energy is converted to electrical energy

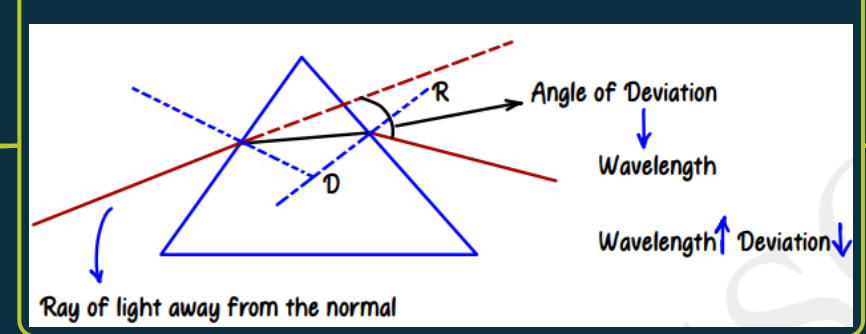
Blind Spot

- Optic nerves meet retina
- No image is formed here

Power of Accommodation

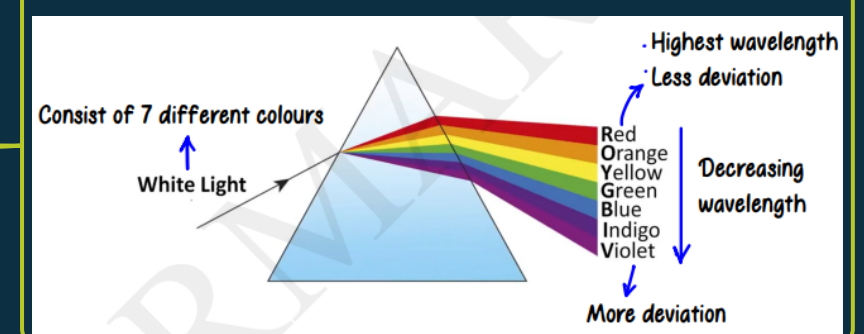
- Eye can change focal length
- help of Ciliary muscles
- Least distance of distinct vision - 25 cm

Refraction of Light



PRISM

Dispersion of White Light





Commercial Unit

220 V, 50 Hz → In electric appliances
Live wire: Red
Neutral: Black
Ground/Earth: Green/Yellow
Colour of wire

1 unit = 1 kWh
 $P \times t$ → 1 kW = 1000 W
1000 x 3600s
 36×10^5 Ws
 3.6×10^6 J → 1 unit
1 hr = 60 mins
60 x 60 = 3600 secs

Electric Power

$$P = VI$$
$$P = I^2 R \quad V = IR$$
$$P = \frac{V^2}{R} \quad \frac{V}{R} = I$$

Materials

Conductor

materials that conduct electricity/allow electric flow through them

eg - usually Metals

Semi-conductor

materials which have conductivity between conductors and non-conductors/insulator

eg - Silicon (usually Metalloids)

Insulator

materials that do not allow electricity to pass through them

eg - usually Non-metals

Heating effect

Applications

Coils in heater made of Nichrome
Alloy of Ni and Cr

Filament made of Tungsten (W)
Has high melting point

Fuse
Low melting point
It is used to protect electrical appliances from excessive current and to prevent short circuits or mismatched loads

The current flowing in a conductor is directly proportional to the voltage across the conductor, provided all the physical condition and temperature remain constant

S.I unit - Ohm

Ohm's Law

By George Simon Ohm in 1827
 $V \propto I$
 $V = IR$
Resistance

Charge

+ve & -ve

Like charges repel each other

Opposite charges attracts each other

S.I unit - Coulomb

$1 e^- = 1.6 \times 10^{19}$ Coulomb

Electric Current

Flow of charge / electrons per unit time

measured by Ammeter

S.I unit - Ampere

Electric Current

$$i = \frac{q}{t}$$

Potential Difference

amount of work done in moving a unit positive charge from one point to other in an electric field

Measured through: Voltmeter

S.I unit: Volt

Potential difference

$$V = \frac{W}{q}$$

Work done to move the charge

Resistance

property of a conductor to resist the flow of charge through it

Factors effecting Resistance

- Length → Length ↑ Resistance ↑ (More collision of electrons)
- Area of cross-section → Area ↑ Resistance ↓ (Less collision of electrons)
- Temperature → Temperature ↑ Resistance ↑
- Nature of material → Increase movement of e^- and K.E

Resistivity/Specific resistance

$$R \propto L$$
$$R \propto \frac{1}{A}$$
$$R \propto \frac{L}{A}$$
$$R = \frac{\rho L}{A}$$
$$\rho = \frac{RA}{L}$$

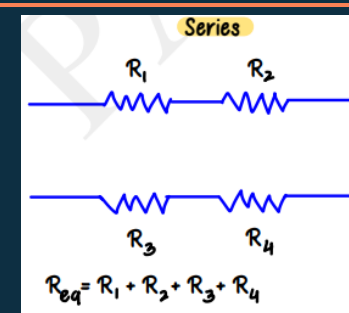
The resistivity of a material is the resistance of a wire of that material

	Material	Resistivity (Ω m)
Conductors	Silver	1.60×10^{-8}
	Copper	1.62×10^{-8}
	Aluminium	2.63×10^{-8}
	Tungsten	5.20×10^{-8}
	Nickel	6.84×10^{-8}
	Iron	10.0×10^{-8}
	Chromium	12.9×10^{-8}
	Mercury	94.0×10^{-8}
	Manganese	1.84×10^{-6}
	Constantan (alloy of Cu and Ni)	49×10^{-6}
Alloys	Manganin (alloy of Cu, Mn and Ni)	44×10^{-6}
	Nichrome (alloy of Ni, Cr, Mn and Fe)	100×10^{-6}
Insulators	Glass	$10^{10} - 10^{14}$
	Hard rubber	$10^{13} - 10^{16}$
	Ebonite	$10^{15} - 10^{17}$
	Diamond	$10^{12} - 10^{13}$
	Paper (dry)	10^{12}

Alloy has greater resistivity than its constituent metals

System of Resistance

same current



different potential difference

Parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Fleming's Left hand rule

Force experienced by the wire $\vec{F}$
 The Magnetic Field $\vec{B}$
 The current in the wire $\vec{I}$

Used when a current carrying conductor is introduced in an external magnetic field

FLEMING'S RIGHT HAND RULE

Generator $\rightarrow$ Mechanical Energy $\rightarrow$ Electrical Energy

Thrust or Motion
 Magnetic Field
 Current

Fleming's Right hand rule

Simple Electromagnet

Iron nail
 coiled wire
 wire
 battery

Soft iron rod that behaves like a magnet when current is passed through it

Current direction upwards

Case 1 $\theta = 90^\circ$
 Case 2 $\theta = 0^\circ$

Magnetic field

Motor works on principle of Fleming's Left Hand Rule
 Electric energy $\rightarrow$ Mechanical energy

Force will be **Maximum**: the angle between the conductor and the magnetic field is 90°
Minimum: The conductor is placed along the direction of magnetic field, whether parallel or antiparallel $\rightarrow$ Zero

COILED WIRE
 DIRECTION OF MOVEMENT
 INDUCED VOLTAGE IN WIRE

It is the phenomenon in which electric current is generated by changing magnetic fields
 Discovered by Michael Faraday in 1831

When the bar magnet is pushed towards the coil, the pointer in the galvanometer deflects

The relative motion between the magnet and the coil is responsible for generation of electric current in the coil

Induction

Electric current passed through the wire

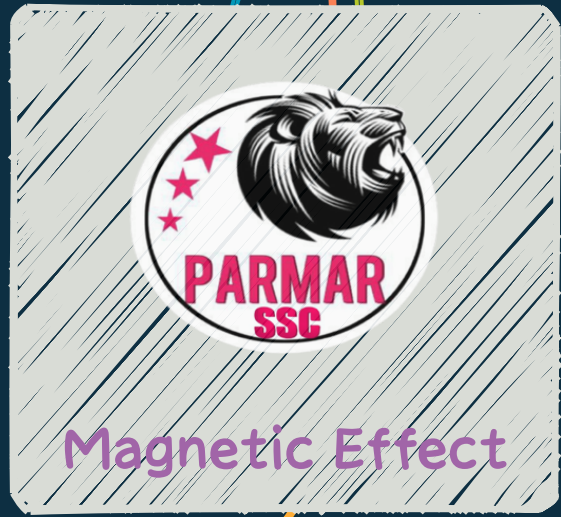
Causing deflection in the compass whenever there is a current in wire showing current carrying wire produces a magnetic field around it

Hans Christian Orsted in 1820

Electromagnetic

Discovery

Magnetic field due to -



Magnetic Field Lines

Magnetic field direction
 Magnetic field lines
 Magnetic field strength

Magnetic field lines originate from North Pole outside the magnet and terminate at South Pole
 Magnetic field line are in the form of closed loop
 Magnetic field lines never intersect each other

Vector quantity
 If they intersect, there will be two directions of magnetic field lines which is not possible

Magnetic field lines around a straight conductor carrying current are concentric circles whose centre lie on the wire

Magnetic Field $\propto \frac{1}{\text{Distance}}$
 Distance $\uparrow \rightarrow$ Magnetic Field $\downarrow$

Current $\uparrow \propto$ Magnetic Field $\uparrow$

Direction of current changes $\rightarrow$ Direction of Magnetic Field changes

current carrying Conductor

Direction of current
 Direction of magnetic field

Maxwells Right Hand Thumb Rule to find direction of magnetic field

Fig 3.9 Magnetic field due to a circular loop carrying current

No. of turns in loop $\uparrow \rightarrow$ Magnetic Field $\uparrow$
 Distance $\uparrow \rightarrow$ Magnetic Field $\downarrow$
 Current $\uparrow \rightarrow$ Magnetic Field $\uparrow$

In centre, the magnetic field lines are parallel and uniform

current carrying Circular Loop

Solenoid

Behaves like a bar magnet when current is passed

In a solenoid the magnetic field is maximum inside the solenoid
 It is uniform

Current passed through a solenoid
 If direction of current is reversed, the direction of magnetic field is also reversed

Magnetic field lines outside the solenoid is very weak $\rightarrow$ Considered to be practically zero

current carrying Solenoid